Acquiring the Features of Negative Indefinites

A comparison of Catalan, Spanish and Italian with West Germanic

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Abstract

Corpus-based and experimental studies show that children acquiring non–Negative Concord (NC) languages often display NC-like behaviour, producing or interpreting sentences with two negatives as a single negation. Previous work has taken this as evidence that children initially posit a non-target NC grammar (e.g., Sano et al., 2009; Thornton et al., 2016; Nicolae & Yatsushiro, 2022; Driemel et al., 2023; Illingworth et al., 2025). The empirical basis for this claim is, however, largely based on Double Negation systems, especially Germanic. This paper presents the first corpus study of NC and negative indefinites in Catalan and Spanish, supplemented by existing and new Italian data. Comparison with Romance and Germanic reveals differences in error rates, age of emergence, and error types, providing evidence against the view that NC is children's 'developmental default'. I argue that these findings are best captured under Zeijlstra's (2004) Agree-based approach to NC.

Keywords— negation, Negative Concord, syntax, L1 acquisition, Romance, Germanic, Catalan, Spanish

1 Introduction

While a significant body of work on the acquisition of sentential negation exists (see Thornton, 2020, for a review), there is comparatively little research on the acquisition of negative indefinites, particularly in Romance languages. Existing work has centred, to a large extent, on English, Germanic and other languages with Double Negation systems (i.a., Jou, 1988; Sano et al., 2009; Tieu, 2013; Thornton et al., 2016; Nicolae & Yatsushiro, 2022; Driemel et al., 2023), with Negative Concord languages having received less attention (though see Tagliani et al., 2022, for an exception in Italian, and White et al., 2022; Biberauer & van Heukelum, 2025a, 2025b, in Afrikaans). At the same time, a range of theoretical analyses have been proposed to account for both NC and DN systems (see Penka, 2020, for a review). Awareness of negation and negative indefinites is reported to emerge relatively early on, though not always in a fully target-like manner (see, e.g., Klima & Bellugi, 1966; Hein et al., 2023, on production; Thornton et al., 2016, on comprehension, and other references above). It still remains a challenge to provide a crosslinguistic account of how the acquisition of negative indefinites differs between NC and DN systems, and how these developmental patterns bear on their proposed formal analyses.

Against this backdrop, this paper has three primary aims: (i) to help fill the empirical gap by providing new data on the production of Negative Concord and Negative Concord Items in Catalan and Spanish children; (ii) to systematically contrast the existing Romance data (Catalan, Spanish and Italian) with the development of negative indefinites in Germanic; finally, (iii) to evaluate the findings obtained in the context of the (synchronic) morphosyntactic analyses of negative indefinites in the literature.

I organise the exposition as follows. Section 2 outlines the theoretical background and identifies a key point of contention in current theories of the acquisition of negative indefinites: whether Negative Concord (NC) is 'simpler' for the learner to acquire than Double Negation (DN), what I will dub the 'NC-default' hypothesis. Section 3 turns to the corpus study. The empirical discussion proceeds in two steps. I first describe the results obtained for the development of NC in Catalan and Spanish, providing the first descriptive overview of NC acquisition in these languages, as well as in Italian. I then compare these patterns with those reported for Germanic languages, drawing also on my own synthesis of the available empirical literature.

First, I show that error rates in Romance production are comparable to those reported for Germanic. However, while the errors are quantitatively similar, they are *qualitatively* distinct: errors of omission of the sentential negator predominate in Romance, whereas Germanic languages display some errors of 'surface' concord. These errors also emerge at *different stages*, with Romance errors appearing much earlier in the acquisition of negative indefinites. I further show that Romance children systematically do not produce errors when the negative indefinite is preverbal (e.g., overproduction of the sentential negator, akin to Strict NC), whereas such errors are attested in English. Second, I gather novel, tentative support for a non-negative analysis of NCIs in Romance. Evidence comes from the presence of 'isolated' negative indefinites at early stages in Germanic but not in Romance: that is, NIs occurring on their own in non-target-like configurations to convey an expressive negative meaning, rather than as fragment answers to a preceding question.

The overall empirical picture militates against strong NC-default accounts (e.g., Thornton et al., 2016; Sano et al., 2009), but also against proposals that attribute the relevant patterns to more sporadic competence deficits (Driemel et al., 2023). Section 4 presents the analysis and discussion. Adopting a parametric approach following Zeijlstra's (2004) Agree-based theory of NC, I argue that the crosslinguistic patterns observed follow best under an account in which NC is derived, rather than the learner's default.

2 Background

This section introduces the theoretical and empirical background of the paper. I summarise Negative Concord and Double Negation languages in §2.1, including some of their theoretical analyses. In §2.2, I provide a synthesis of the literature thus far on the acquisition of negative indefinites, identifying the open questions to which this paper contributes.

2.1 Negative Concord and Double Negation

2.1.1 Typological overview

Languages are known to vary in how they encode negative dependencies and negative indefinites. Broadly, languages can be split into *Double Negation* (DN) and *Negative Concord* (NC) languages, with a further subdivision existing within the latter (e.g., de Swart & Sag, 2002). DN languages, such as (standard) English, German and Dutch, require negative indefinites (*nothing*, *nobody*, etc.), henceforth NIs, to appear on their own, without overt expression of sentential negator or other negative markers. Examples in (1) illustrate sentences with single sentential negation readings.[1]

(1) Double Negation or Non-Negative Concord languages

a. English

I have seen **nobody** today.

b. German

Sie hat **nichts** gekauft.
she have.3SG nothing buy.PST.PTCP
'She has bought nothing.'

c. Dutch

Geen enkele deelnemer had deze prestatie verwacht.
no single attendee have.PST.3SG that performance expect.PST.PTCP
'No a single attendee expected that peformance.'

NC languages, on the other hand, require a negative marker to co-occur with negative indefinites, in at least some contexts. Two types of NC languages are generally distinguished: *strict* and *non-strict* NC languages. These diverge in the extent to which the negative marker has to be present alongside the NI; I term negative indefinites in NC languages *Negative Concord Items* (NCIs), following the literature (see discussion in, e.g., Zeijlstra, 2004). In strict NC, the negator is found in all contexts along with the NCI (e.g., both postverbal and preverbal NCIs). Examples of strict NC languages are Greek, Hebrew, Turkish, Romanian, Russian, among several others.

(2) Strict NC (Czech)

a. **Nikdo ne**-volá.
nobody NEG-call
'Nobody calls.'

b. **Ne**-volá **nikdo**.
NEG-call nobody
'Nobody calls.'

(Zeijlstra, 2004: 214, 251)

[1]Unless otherwise noted, the examples are mine. For languages other than English, Spanish or Catalan, examples have been double-checked with native speakers.

Non-strict NC languages, in contrast, only sanction the sentential negator when it *precedes* an NCI; that is, the NCI must be postverbal when the negator is overt. Catalan and Spanish are two non-strict NC languages, and illustrate the patterns below. Italian, Portuguese and West Flemish also fall into this typology.

(3) Non-strict NC in Spanish and Catalan

a. Negative doubling (Spanish)

*(**No**)	vino	**nadie**.
not	come.PST.3SG	n-body

'Nobody came.'

b. No negator with pre-verbal NCIs

Nadie	(***no**)	vino.
n-body	not	come.PST.3SG

'Nobody came.'

c. Optional negator with pre-verbal NCIs (Catalan)

Ningú	(**no**)	va	venir.
n-body	not	AUX.PST.3SG	come.INF

'Nobody came.'

The position of Catalan within this category is, however, not clear-cut: it differs from Spanish in having further microparametric variation. It optionally allows the negative marker to co-occur with preverbal NCIs (3c), making it more similar to strict NC languages in this respect. We set this aside here, and continue treating Catalan as largely non-strict following the literature (but see §3, for brief discussion).

The sentential negator is not the only negative element or context that can license NCIs in NC languages. In non-strict NC, combinations of an NCI plus another NCI are grammatical and lead to single negation readings, so-called 'negative spread' (4). These examples lead to double negation readings in (Standard) English (5), in contrast.[2]

[2]Whenever I refer to 'English' in broad terms, this should be understood as 'Standard English'. Negative Concord is widespread across many English varieties, particularly in the UK and in African American English (Tubau, 2008; Blanchette, 2015). As such, the classification of English within the DN/NC typology must be made at the level of individual varieties, rather than the language as a whole. Moreover, even the status of Standard English remains debated. Some accounts argue that it constitutes a 'hybrid' system, potentially licensing both DN and NC in some contexts even in standard-adhering varieties (see Zeijlstra, 2004; Tubau, 2008; Blanchette, 2015). I return to this issue in §4.

(4) Negative spread

a. Czech

Dnes **nikdo** **ne**-volá **nikoho**.
today nobody NEG-call nobody
'Today nobody calls anybody.'

b. Catalan

Ningú va dir **res**.
nobody go.AUX.PST.3SG say.INF nothing
'Nobody said anything.'

(Giannakidou & Zeijlstra, 2017: 9)

(5) Double Negation

a. I **didn't** want **nothing** (≈ I wanted something/anything).

b. **Nobody** said **nothing** (≈ Someone said something).

Finally, NCIs can also surface as (negative) responses to a previous wh-question, so-called Negative Fragment Answers (NFAs) without need for an overt negator, as in Catalan *Qui va venir?* ***Ningú*** 'Who came? Nobody'. Other relevant negative contexts (antiadditive, antiveridical contexts; see review in Zeijlstra, 2022) are *without*-clauses and neg-raising contexts, which are known to license negative polarity items, a broader class encompassing NCIs. This paper will largely be concerned with constructions featuring the sentential negator *no* in Catalan and Spanish or with Negative Fragment Answers (NFAs); other antiveridical contexts are rare or unattested in early child speech.

2.1.2 (Some) theoretical analyses of Double Negation and Negative Concord languages

A range of theoretical analyses have been proposed to capture the differences between DN and NC languages. A full review of them is beyond the scope of this paper; I limit myself to illustrating the key theoretical points of contention that have also been taken up in language acquisition theories (the latter introduced in the next section, §2.2).

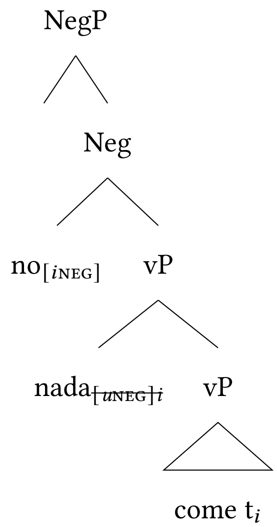
An arguably standard assumption across approaches is that negative indefinites in DN languages like English contribute the semantic negation reading on their own. Addition of further negative markers then accordingly triggers double negation readings, cancelling out negation into a positive reading. A common analysis has thus been to analyse negative indefinites in these language as negative quantifiers: they are inherently semantically negative, specifically their lexical entries contain a negation operator as well as an existential meaning component (Montague, 1973; Barwise & Cooper, 1981).[3]

The analysis of NC, conversely, is a matter of significant theoretical debate. The central bone of contention lies in whether NCIs are inherently semantically negative or whether the negative semantics comes about by some other means (the sentential negator, or covert negative operators in the case of non-strict languages). A proponent of the former position is Zanuttini (1991), and subsequent work (i.a., Haegeman & Zanuttini, 1991; Haegeman, 1995; de Swart & Sag, 2002), which argues for a negative quantifier analysis of NCIs (akin to English). Simplifying slightly, the proposal is that NCIs must be located in NegP for them to be interpreted semantically as the sentential negator, e.g., via Neg-Absorption of two negative markers/quantifiers into a single negative semantic component (see Haegeman & Zanuttini, 1991; Haegeman, 1995). Zeijlstra (2004), in opposition, has argued for a non-negative semantics of NCIs (as in earlier work, Ladusaw, 1992; Giannakidou, 2000). He likens NC to syntactic agreement, implemented by upward Agree (cf. Chomsky, 2000, 2001). NCIs themselves are non-negative, endowed with an uninterpretable [uNeg]; as such, they require 'licensing' by an overt sentential negator or by a covert negative operator ([iNeg]) (in cases where the negator is not present, such as preverbally in non-strict NC or in negative fragment answers). [uNeg] can then Agree upward with [iNeg] (24). A very schematic tree-representation is given below for Spanish *no come nada* ('he/she doesn't eat anything'):[4]

[3]In logical formula, this can be spelled out and decomposed as follows: $[\![nobody]\!] = \lambda P.\neg\exists x[person(x) \,\&\, P(x)]$.

[4]I gloss over the raising of NCIs for simplicity, but their movement is key in both Zanuttini (1991) and Zeijlstra (2004).

(6)



In the absence of space for a full discussion, I refer the reader to Penka (2020), Driemel et al. (2023: ch. 3), and Tubau et al. (2024) for detailed reviews. The brief exposition provided here is intended only to illustrate that the analysis of NCIs remains contested. As such, the NC-default hypothesis is one that needs to be considered in the context of the various existing theories of NC, since a NC-default does not straightforwardly follow from all existing synchronic analyses. Moreover, Zeijlstra's (2004) theory makes independent developmental predictions, which we elaborate in §4.

2.2 Acquisition of negative indefinites

Research on the acquisition of negative indefinites remains primarily limited to Germanic and other Double Negation languages (e.g., Japanese), with the exception of Tagliani et al.'s (2022) study on Italian and White et al. (2022), and Biberauer & van Heukelum (2025a,b) on Afrikaans. The Germanic data from English, German and Dutch has emphasised, to different extents, (non-target-like) single negation interpretations in children's comprehension of Double Negation constructions, and the existence of Negative Concord 'errors' in their production data.

Thornton et al. (2016) found that English children (age range 3;6–5;8) strongly favour a single negation interpretation of sentences, in contrast to adult participants: when the double negation interpretation made the test sentences false, children nonetheless judged them to be true, unlike adults (see also Thornton & Tesan, 2013). Conversely, when it rendered them true, children again incorrectly interpreted them to be false. Thornton et al. interpret the data as supporting the theoretical prediction that children acquiring (Standard) English will entertain a grammatical stage with a NC grammar (following Zeijlstra, 2004, in assuming English is 'hybrid' and can show NC behaviour; see §4 for further discussion). This NC-like comprehension stage is also corroborated by Nicolae & Yatsushiro (2022) for German (4;2-6;5), by a range of work on Mandarin and Japanese (Jou, 1988; Zhou et al., 2014; Hu et al., 2024; Sano et al., 2009), and by Maldonado & Culbertson (2021) for artificial language learning, where participants more readily induce NC grammars than DN grammars. Finally, Illingworth et al. (2025) show that the Negative Polarity Item *any* in English is frequently misinterpreted as the negative quantifier *no* in children aged between 2;00 and 6;00, which they argue is compatible with learners inducing an NCI analysis for *any*.

Hein et al. (2023) and Driemel et al. (2023) make a related proposal adducing new production data from English, German and Dutch children. A range of NC-like errors are observed in their production; specifically, they occasionally overtly produce the sentential negator alongside negative quantifiers like *nothing*. For example, these 'errors' involve utterances of the sort *I* ***don't*** *see* ***nobody*** (Adam, 3;10.15), and *En Rosa mag* ***niet geen*** *spelletje*, lit. 'And Rosa may not play no game'. Their dataset is illustrated in Table 1 and Figure 1 and is described more in depth in §3.4. Note that the child Adam is set aside in the table and figure as he is exposed to African American English, which features Negative Concord.

Table 1: Production of NIs and NC across English, German and Dutch (Driemel et al., 2023: 8). Proportion = overall mean error rate across files. Peaks = the highest error rate recorded.

	Utterance Count				
Language	**Total**	**with NI**	**with NC**	**Proportion (NC/NI)**	**Peak(s)**
English (all)	328,972	909	184	20.2% (Hein et al., 2023)	34%
English (w/o Adam)	283,399	666	53	8.0%	7.1% & 13.4%
German	338,407	2665	45	1.7% (Hein et al., 2023)	5.9%
Dutch	220,617	857	6	0.7%	3.7%

It remains unclear, nonetheless, what causes these target-input divergences and, in particular, how the asymmetry in the amount of NC productions vs. comprehensions is to be explained. A common account has been to attribute NC errors to induction of a genuine NC grammar (Thornton

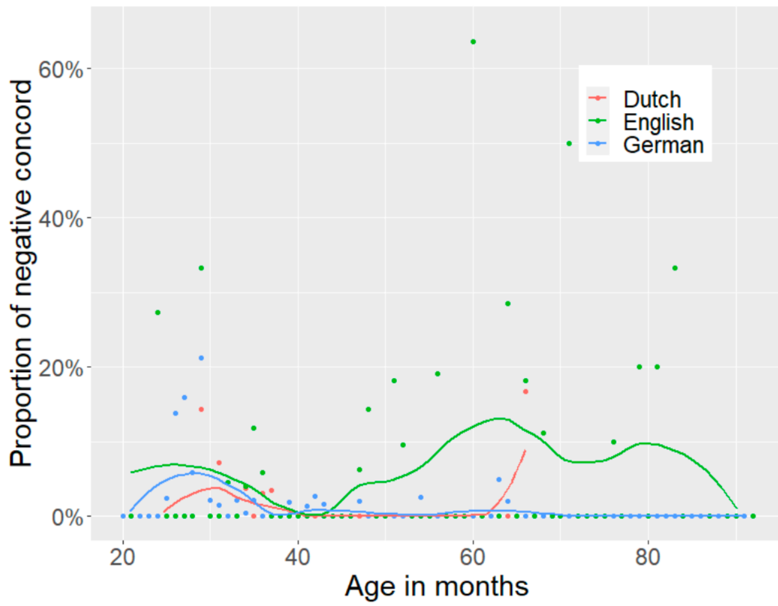


Figure 1: Proportion of NC over age in Dutch, English (excluding Adam) and German (Driemel et al., 2023: 8).

et al., 2016; Sano et al., 2009),[5] or to some NC-like stage (Driemel et al., 2023),[6] all variously implemented. The strongest hypothesis – that children's non-target-like performance is due to a NC grammar – faces some immediate issues: it does not explain why errors in comprehension are far more common than in production (the reverse of what is generally, though not always, observed in syntactic development). If Germanic children have developed a NC grammar, NC ought to be at least very widespread in both their production and comprehension.

The explanation given by Hein et al. (2023) and Driemel et al. (2023) is more nuanced, and aims to partly dissociate the root(s) of the patterns in production vs. comprehension. Hein et al. (2023) acknowledge that a 'true' NC-stage in children's grammars is unlikely, and attribute the quantitative differences between German and English in their paper (part of Table 5) to independent properties of the languages, namely the presence of a rich and regular system of Negative Polarity Items in English. The relatively free word-order of German, in contrast to English, is argued to

[5] Crucially, however, it is not specified what this 'NC default' here should be interpreted to involve, namely whether we should expect strict or non-strict NC, or either, and why.

[6] 'NC-like' here need not equate with 'NC grammar', rather with a system that can produce outputs that *resemble* NC-grammars, even if underlying it is not (fully) a NC system. We return to this with the discussion of Driemel et al. (2023).

also explain the lower proportion of errors in German; preverbal negative indefinites are found to be harder and more error-prone in both German and English, but this configuration is most easily avoided in German, by resorting to postverbal subjects (the latter allowed in German due to its V2 rule, but generally disallowed in Standard English).

Driemel et al. (2023) then take this further with Dutch data, which likewise presents minor error rates, and propose to understand the production patterns under a grammatically-mediated analysis, following the Meaning First model (Sauerland & Alexiadou, 2020; Guasti et al., 2023). In this model, semantic (not syntactic) structure is primitive and it is composed into a so-called conceptual structure, which precedes externalisation. Conceptual structure is the first step of grammatical composition. Externalisation then maps ('compresses') the conceptual structure into an array of morphemes/exponents. Under this approach, the proposal is that children will prefer one-to-one mappings between semantics and PF exponents, following earlier work on the so-called Transparency Principle (Slobin & Bever, 1982; Jackendoff, 1990; Weist et al., 1997; Van Hout, 1998).

Their system proceeds as follows. Simplifying somewhat, NC is seen as a system that introduces a 'redundancy' in the assumed semantic structure, namely two NEG operators, both of which are spelled out. This redundancy is proposed to come about through a rule-based process of NEG-duplication, given in (7a).

(7) *Compressor rules that disrupt one-to-one mapping*

a. NEG-duplication: $\varnothing \longrightarrow$ NEG / NEG [_ $\exists$

b. NEG-deletion: NEG $\longrightarrow \varnothing$ / _ [NEG $\exists$

(Driemel et al., 2023: 24)

NEG-duplication is seen as a compulsory enrichment rule (in the sense of Distributed Morphology) that duplicates NEG in the context of an existential quantifier ($\exists$), leading to extended exponence (Negative Concord). The original NEG operator is semantically negative, while <NEG> is duplicated at the morphological component, and therefore has no semantic import. NEG is spelled out as the negator, and <NEG>+$\exists f$ as the NCI/NI. Crucially, NEG-duplication is assumed to be in the grammar of non-NC systems too. Therefore, non-NC grammars must not overtly realise one NEG operator through NEG-deletion (an impoverishment rule, see (b) above).[7] NC grammars

[7] See Driemel et al. (2023) for additional details on this implementation, including the role of an additional operation

therefore have one fewer rule in the morphological component, whilst DN grammars must order NEG-deletion *after* NEG-duplication:

(8) *Rule orders for NC and DN systems*

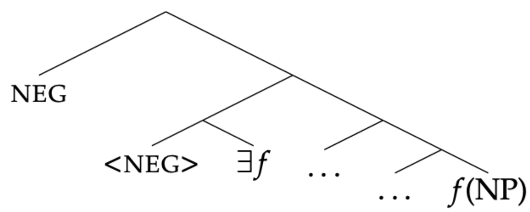
a. NC grammar: NEG-duplication (see 9a)

b. non-NC grammar: NEG-duplication $\prec$ NEG-deletion (see 9b)

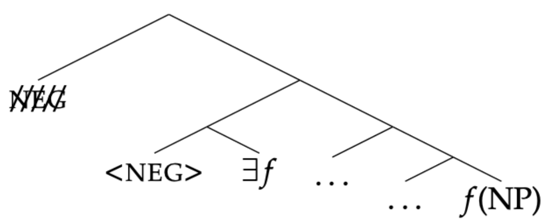
(Driemel et al., 2023: 36)

(9) illustrates the derivational outputs of this rule system, for both NC and non-NC.

(9) a. *NC grammar*: NEG-duplication



b. *Non-NC grammar*: NEG-duplication $\prec$ NEG-deletion



(Driemel et al., 2023: 24ff)

Errors of 'undercompression' – i.e., of overt production of material that should be deleted – are expected, whereupon the approach predicts Negative Concord errors in children acquiring Double Negation languages given the (compulsory) duplicated <NEG>+$\exists f$. NC here is, then, again some sort of rule 'default', in this case at the mapping between conceptual and phonological form (traditionally, Logical Form, LF, and Phonological Form, PF). Double Negation systems require more derivational work to be generated, because the redundant NEG operator must be jettisoned;

(bundling) which I set aside here in the interests of brevity.

failure to do so leads to undercompression. They propose that additional factors beyond their non-target grammar (e.g., higher processing load) amplify the amount of NC in comprehension, helping to explain the asymmetry observed with production. Arguably, however, one could still ask why a non-target grammar at the level of LF/PF would manifest itself so sparingly in the production data (Table 5), but potentially does to a much greater extent in comprehension. It likewise remains unclear what motivates the inclusion of NEG-duplication in both NC and non-NC grammars.

To summarise, all approaches introduced thus far predict 'errors' with negative indefinites to predominate in Double Negation languages, since NC systems are presumably grammatically more 'basic'. This is an explicit empirical prediction to be tested in §3. I dub these approaches 'NC-default' approaches for brevity; though, as noted, their implementations vary significantly.

Some questions still stand: is the small proportion of errors in Germanic production sufficient to warrant a *grammatical* or *competence*-based analysis of them, as emphasised by Hein et al. (2023)? Is the empirical data, across both Germanic and also NC languages (e.g., Romance), sufficient to justify an approach where NC grammars are, in a way, more 'default', either developmentally and/or synchronically? For one, Tagliani et al. (2022) also report errors in their Italian production data, suggesting that occasional production errors are not exclusive to Germanic NIs. Double Negation is scarce in the input and hard to process (Horn, 1989) – this holds even in Negative Concord languages, which allow double negation readings in some fragment answers (Moscati, 2020).

As noted, it remains unclear whether very low error rates can justify non-target-like representations; this is ultimately a question of whether occasional non-target-like productions (e.g., as sporadic performance 'slips') would be independently expected in children who have converged on a (mostly) adult-like system. It is also equally possible that the existing uncommon errors are nevertheless systematic and conditioned, which likewise requires exploration. Additionally, not all synchronic approaches to NC analyse NC as more 'fundamental': Zeijlstra (2004, *et seq.*) has notably argued that Double Negation languages are actually the 'default'; negative indefinites, being negative quantifiers, do not require a [uNeg] to be postulated, their negative semantics is inherent. NC, on the other hand, requires a [uNeg] on indefinites to derive and license their non-semantically negative nature. Zeijlstra and other approaches (e.g., Deal, 2022) take NC to be derivative (with DN being syntactically less complex), suggesting that the crux of the developmental questions at stake above is (at the very least) theoretically debated.

In this paper, I contribute to this literature by investigating the acquisition of Negative Concord and negative indefinites in three Romance languages – two thus-far unstudied languages, Catalan and Spanish, supplemented with Italian data (both already-existing and newly collected). I then compare their acquisition, and error frequency, with Germanic data. Besides providing a first description of the acquisition of NC in these languages, I show that this comparison sheds light on the plausibility of a NC stage in Germanic and crucially highlights children's early awareness of the typological properties of their L1 in both language families.

3 The corpus study

3.1 Methodology

We selected all available longitudinal CHILDES corpora for Catalan and Spanish, both monolinguals and bilinguals, that contained NIs. The Catalan data comes from 14 children, 9 monolinguals and 5 bilinguals (Catalan-Spanish bilinguals only), spanning ages 1;01-4;03 (corpora Serra-Solé, Serra & Solé, 1986–1989; Jordina, Llinàs-Grau, 2000; Júlia, Bel, 1998; EvaMireiaPascual, Llinàs-Grau, 1997; Vila, Vila, 1990). For Spanish, we collected data from 20 children: 9 monolinguals and 11 bilinguals (either Spanish-German or Spanish-English bilinguals).[8] Similarly, the age-range was 0;11-5;10 and the data derived from the following corpora: Aguirre (Aguirre, 2000), Linaza (Linaza et al., 1981), Marrero (Cappelli et al., 1994), Montes (Montes, 1987), Aguado-Orea-Pine (Aguado-Orea & Pine, 2015), Ornat (López Ornat, 1994), Remedi (Remedi, 2014), Vila (Vila, 1990), Llinàs-Ojea (Llinàs-Grau & Ojea López, 2005), Pérez (Pérez-Bazán, 2002), PhonBLA (Lleó et al., 2003), FerFurLice (Liceras et al., 2008).

The corpus study quantifies the development of NCIs and their use in Negative Concord constructions in the two languages. We coded for use of NCIs (*nothing*, *nobody*, *never*, *no*+DP, and the focus particle *ni* 'not even'[9]); for the structure in which the indefinite was found (Negative Concord/Doubling, Negative Fragment Answer, Error); for position in the utterance (pre/postverbal, if applicable); and for argumental status (External Argument or EA, Internal Argument or IA, Ad-

[8]The type of bilingual data collected is necessarily unbalanced and heterogenous (Catalan-Spanish, Spanish-German and Spanish-English). All relevant Spanish/Catalan corpora in CHILDES were considered, so this is simply due to lack of availability of more balanced language-pairs in CHILDES.

[9]*Ni* in Catalan and Spanish behaves as an NCI insofar as it is only licensed under negation (Espinal & Llop, 2022).

junct, if applicable)[10]. The NCIs investigated and the coding is summarised in Table 2. We discarded potentially routine constructions and repetitions of adult utterances.

The Catalan and Spanish data – thus far undescribed – are the focus of this corpus study. Additionally, however, we also collected data for 8 children in 3 Italian monolingual corpora (Antelmi, Antelmi, 1997; Calambrone, Cipriani et al., 1989; Tonelli, Tonelli et al., 1995), to compare their development (already studied in Tagliani et al., 2022) to the Ibero-Romance children here. The Italian data will be referred to sparsely, but will be used to contrast homogeneity (or lack thereof) in the emergence of NCIs across (some) Romance languages. Table 2 summarises the data collected:

Table 2: NCI items and syntactic configurations studied

	Catalan	**Spanish**	**Italian**
NCIs	*Res* 'nothing'	*Nada* 'nothing'	*Niente* 'nothing'
	Ningú 'nobody'	*Nadie* 'nobody'	*Nessuno* 'nobody'
	Mai 'never'	*Nunca* 'never'	*Mai* 'never'
	Cap 'no+DP'	*Ningún/a/os/as* 'no+DP'	*Nessun/a* 'no+DP'
	Ni 'not even' (focus particle)	*Ni* 'not even' (focus particle)	–
Structures with NCIs	NC structures: e.g., Sp. *no quiero nada* ('I don't want anything')		
	Negative Fragment Answers (NFAs): as a response to a wh-question		
	'Errors': e.g., omission of negator with postverbal NCIs (Sp. **quiero nada*) or production of sentential negator with preverbal NCIs (Sp. **Nadie no vino*; only grammatical in Catalan)		
Placement	- Preverbal		
	- Postverbal		
	- N/A		

[10]We coded subjects of existential and copular constructions separately.

Argument structure	- Internal Argument (IA) - External Argument (EA) - Adjunct - NCI preceded/followed by copula - NCI preceded/followed by existential verb - N/A

Since the corpus data sizes necessarily vary across Catalan and Spanish (with fewer corpora available for the former), we calculated the number of utterances per file and per corpus with CLAN, to be used for later normalisation of the counts obtained. Overall, we obtained 471 utterances with an NCI in Catalan and Spanish; 550 in total, including Italian. It is also worth noting that both monolingual and bilingual datasets were included, to maximise the sample size (especially for Catalan). In this paper, I will generally not differentiate between the results for monolinguals vs. bilinguals, on the grounds that few to no differences are observed (when they do potentially differ, this will be pointed out). I leave it to a future task to obtain a more complete picture of how NC is acquired in Romance bilinguals, particularly those acquiring a NC and a non-NC language.

This section introduces the empirical data as follows: §3.2 focuses on the primary contribution of the paper, the results of the corpus study on Catalan and Spanish negative indefinites, with supplementary data from Italian. This subsection is primarily descriptive. §3.4 then compares the outcomes obtained with the data reported for Germanic and provides a synthesis of where and how the two language families diverge, if at all, and how this fares with existing approaches to the acquisition of NIs.

3.2 Results

Table 3 provides an overview of the occurrences of NCIs found in the dataset, and their structure (as part of full NC structures, as Negative Fragments or NFAs, or as Errors). The data is reported twice (Table 3), split by Language first, and then by Acquirer Type (Monolingual vs. Bilingual; Table 4):[11]

[11]The identical numbers for Post-V Errors in both Table 3 and Table 4 are coincidental, they do not correspond to the same tokens.

	Error		NC			
Language	Pre-V	Post-V	Pre-V	Post-V	**NFA**	**Total**
Catalan	0	0	5 (9.3%)	49 (90.7%)	32	86
Spanish	0	15 (100%)	39 (16.4%)	199 (83.6%)	131	384
Italian	0	11 (100%)	1 (2.4%)	42 (97.6%)	26	80
Total	0	26 (100%)	45 (13.4%)	290 (86.5%)	189	550

Table 3: Counts for Error, NC and NFA by language

	Error		NC			
Acquirer Type	Pre-V	Post-V	Pre-V	Post-Vo	**NFA**	**Total**
Monolingual	0	15 (100%)	22 (13.2%)	145 (86.8%)	120	302
Bilingual	0	11 (100%)	23 (13.7%)	145 (86.3%)	69	248
Total	0	26 (100%)	45 (13.4%)	290 (86.5%)	189	550

Table 4: Counts for Error, NC, and NFA counts by child group

Out of the structures considered (NC, NFA, Error), NC structures are the most common. These show a skew towards *postverbal* uses of NCIs (86.5%), only 45 instances of preverbal NCIs are found across the three languages. Monolingual vs. Bilingual children are also highly symmetric in their uses of NCIs, as can be seen by the near-identical proportions per cell in Table 4. Significantly, too, errors are found *exclusively* in postverbal contexts. In other words, only errors of omission of the compulsory preverbal negator are found; errors of overproduction of the negator when the NCI is preverbal are *not* attested. I return to postverbal/preverbal orders below and in §4.

The overall frequency evolution of NCI use across the data is pictured below. All figures below are normalised by utterance count in each file, so as to be comparable across files and corpora.

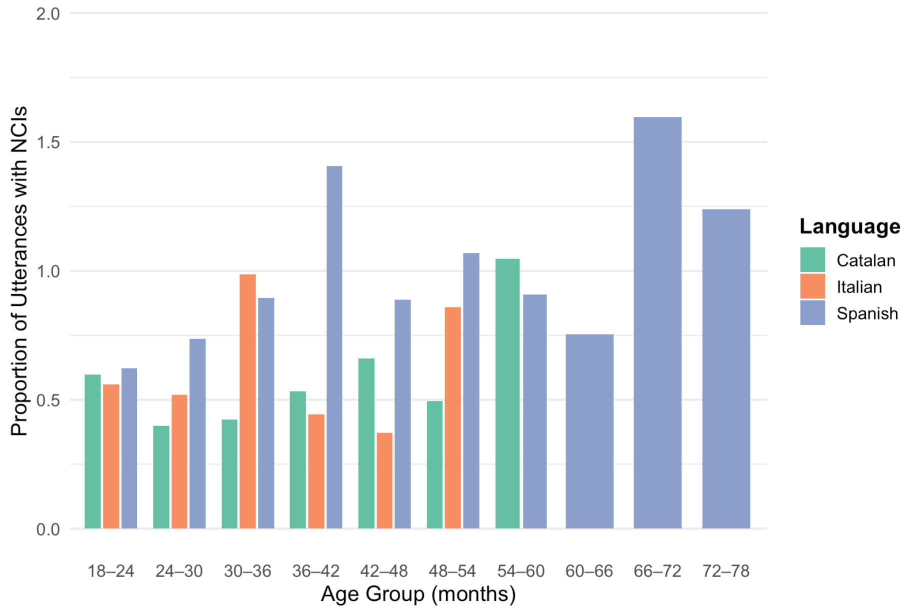


Figure 2: Use of NCIs by age-group and language

As reported for Italian by Tagliani et al. (2022), sentences containing an NCI emerge between 24-36 months, generally around month 30 (full range 21-38 months). NFAs are produced at around the same time, or in some cases, later.[12] The onset age of uses of NCIs in by structure type (NC construction, NFA, Error) is plotted in Figure 3 and Figure 4. Bilingual emergence timings are generally later relative to monolinguals, but they remain broadly comparable.[13]

[12]In Catalan, NFAs emerge later. I do not draw any conclusions from this trend: Catalan is the language for which fewer corpora are available.

[13]I comment on the seemingly later emergence of errors in bilinguals in footnote 17 in §3.4.1, once the full data on errors is introduced. At this point, it is nonetheless worth noting that both box-plots for Error in Figure 4 rely on a very small sample (only 5 monolinguals and 6 bilinguals produce errors), meaning the divergence could be accidental. The same is true for the monolingual age results for Errors.

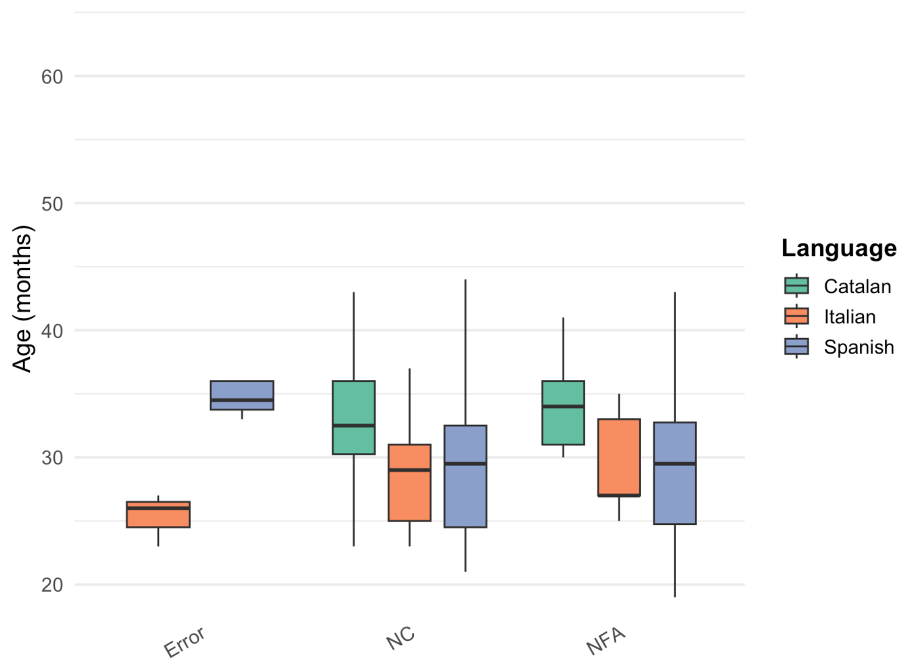


Figure 3: Age of emergence by Structure type and Language

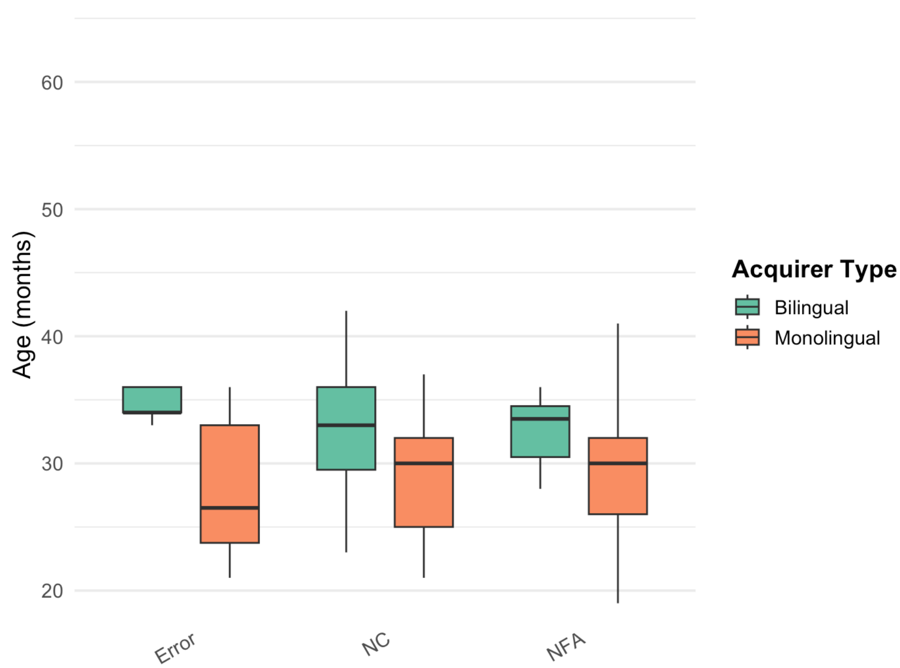


Figure 4: Age of emergence by Structure type and Acquirer type

I use as ‘age of emergence’ the *first* use of an NCI in a grammatical context. This is clearly a very liberal metric, with well-known issues (see, e.g., discussion in Villa-García, 2014); however, I refrain from adopting an emergence metric based on repeated uses of the relevant structure (Snyder, 2007), as the vast majority of children do not have many tokens of NCIs. This would unhelpfully bias estimates to significantly later ages (this is especially true for errors, which are very rarely produced and often contain one token per child only). Any comparisons I will draw with other data outside Romance will also follow this metric, to allow for comparison. Future work with bigger samples should corroborate the ages of emergence cited here.

The first NCI to be produced is almost invariably *nothing* (Figure 5), while the other NCIs show more variation in age of emergence. The focus particle *ni* is sparsely produced, and shows more variation across Catalan and Spanish.

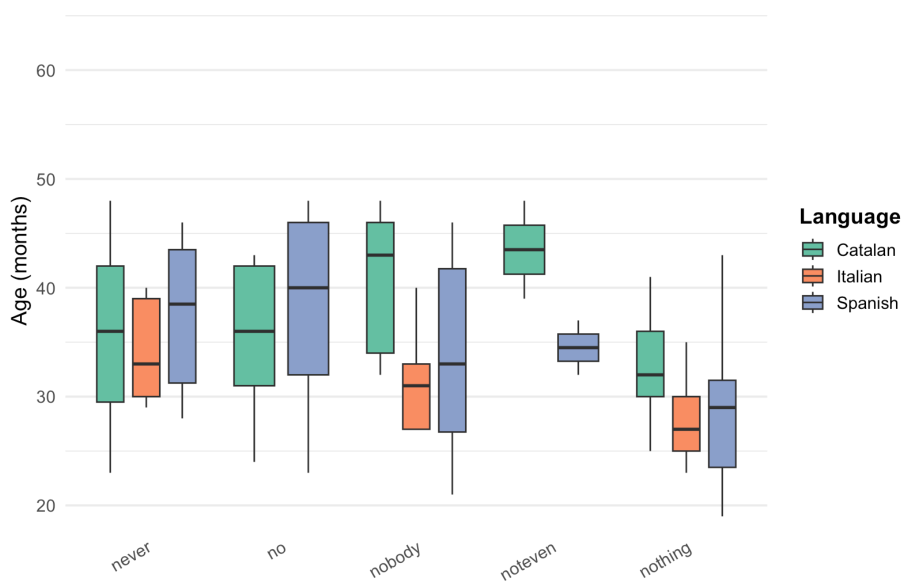


Figure 5: Age of emergence by NCI and Language

When NC structures emerge, they are generally target-like, as shown below, bar some occasional errors, which we discuss in the following section.

(10) Catalan, Gisela (2;08, 2.6 MLUw)

I ara **no** vindrà **ningú**.
and now not come.FUT.3SG nobody

‘And now no one will come.’

(11) Laura (3;05, 2.55 MLUw)

No, que **no** fa **res**.
no that not do.3SG nothing

‘No, because he doesn’t do anything.’

(12) Spanish, Juan (2;03.10, 2.38 MLUw)

No he comido **nada**.
not have.3SG eat.PST.PTCP nothing

‘I haven’t eaten anything.’

(13) Irene (2;05.27, 3.99 MLUw)

Y estaba durmiendo y **nadie** la podía mole(s)tala.
and be.PST.IPFV sleeping and nobody CL.DO= can.PST.IPFV annoy.INF=CL.DO

‘And she was sleeping and nobody could annoy her.’

(14) Italian, Elisa (2;01.19, 4.38 MLUw)

Non fa **niente** il bambino.
not do.3SG nothing the boy

‘The boy isn’t doing anything.’

(15) Marco (2;00.00, 1.89 MLUw)

Non c’è **niente** qua.
not CL.LOC=is nothing there

‘There is nothing there.’

We now report on the interaction between NCI and word-order and argument frames. We observe that postverbal orders predominate across all kinds of NCIs, with the exception of the adjunct *never* (Figure 6). Contrast this with Figure 7, which sheds light on the interplay between order and argumental structure in Catalan and Spanish: *nothing* and *no*+DP are primarily used as Internal Arguments, and so by implication, almost always postverbally. Only *nobody* is predominantly used as an External Argument. Nonetheless, *nobody* subjects are still preferentially found postverbally (Figure 6) – a pattern that is well-reported for both (adult) Catalan and Spanish (i.a., Belletti, 2004; Ortega-Santos, 2008; Etxepare & Uribe-Etxebarria, 2008; Forcadell, 2013), and Germanic children (Bill et al., 2024, 2025).

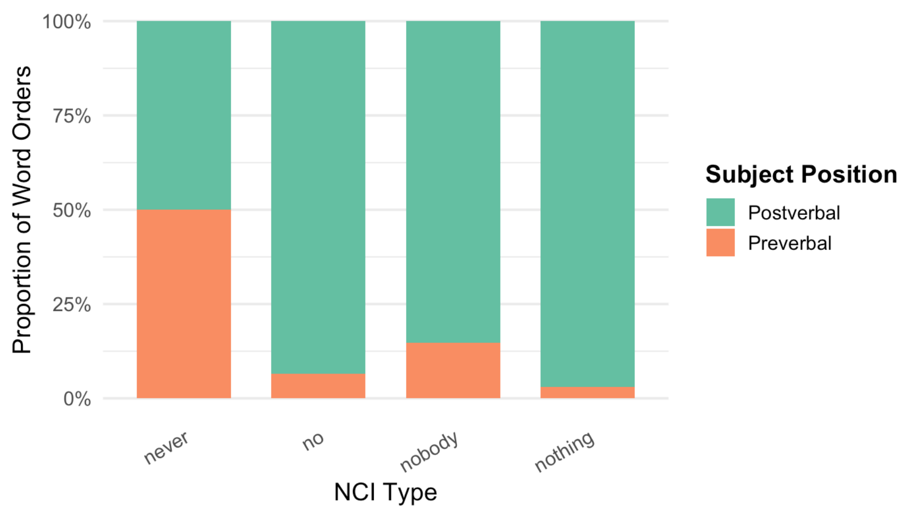


Figure 6: Proportion of Word Orders by NCI

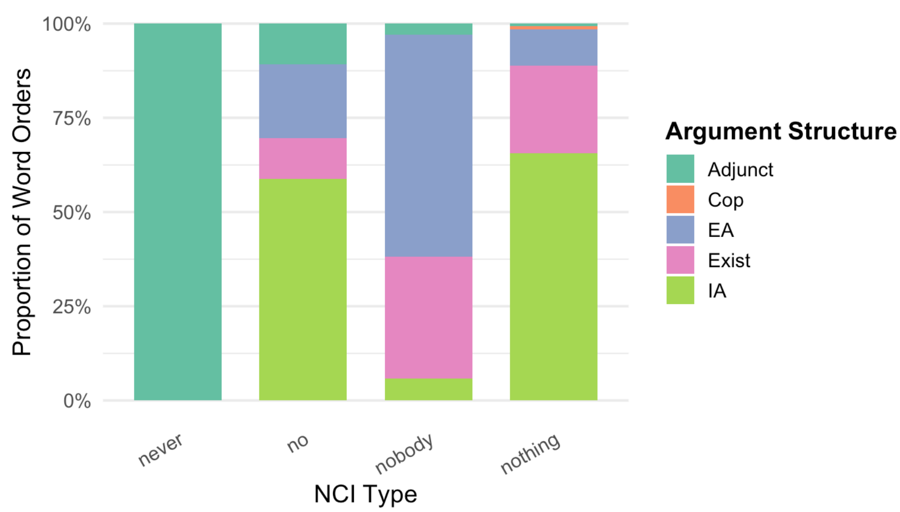


Figure 7: Proportion of Argument Structure Frames by NCI

Finally, I discuss error types in the children's production data. Only one type of error is observed in the Romance data: errors of *omission* of the negator with postverbal NCIs. Errors are generally found early on in the children's production data (mean 32.2 months, cf. 30 for NC). They are infrequent and, in fact, entirely unattested in the Catalan sample (Table 4).

(16) a. Spanish, Magín (1;09.27; MLUw 1.71)

E(s) **nadie**.
is nobody

'He/she/it is nobody.'

b. Carla (3;00.16; MLUw 5.34)

Decía **nada**.
say.PST.IPFV nothing

'He/she/it said nothing.'

c. Italian, Rosa (2;07.26, 2.84 MLUw)

Questo fa **nulla**, guarda.
this do.3SG nothing look.IMP

'This is doing nothing, look.'

As noted above, another logically possible error would be one where negation is incorrectly produced (in Spanish and Italian) with preverbal NCIs, much like strict NC languages. These errors are unattested in our dataset, pointing to two conclusions: the errors attested indicate some initial struggles in overtly producing the overt negator, leading to early productions that are analogous to structures found in DN languages ('DN-like' errors). The reverse pattern ('strict NC'-like errors) is absent, suggesting that the Romance children are nonetheless aware of the non-strict nature of the languages at the point in which NC structures start being produced.

Finally, the corpus data proves too limited to test one important aspect of the development of Catalan NC: the acquisition of the optionality of sentential negation with preverbal NCIs (§2). Since the majority of the NCI tokens are postverbal, it is at present not possible to diagnose which structure (with/without the negator) emerges first, if any. Only 3 examples exist with preverbal NCIs in the database, all without the sentential negator (17). This sample is clearly too small to draw any conclusions.

(17) a. Guillem (3;11.20, 2.85 MLUw)

Res li he dit.
nothing CL.IO= have.1SG say.PST.PTCP

'I've said NOTHING to him/her.'

b. Laura (4;00.10, 3.18 MLUw)

Res he menjat.
nothing have.1SG eat.PST.PTCP

‘I have eaten NOTHING.’

c. Caterina (4;03.21, 3.84 MLUw)

És	que	jo	**mai**	he	vist	porquets	a	la	casa	del	Ramón.
is	that	I	never	have.1SG	see.PST.PTCP	piglets	at	the	house	of-the	Ramón

‘I’ve never seen piglets at Ramón’s house.’

3.3 Interim summary

Overall, then, the Romance picture from Catalan, Spanish and Italian shows a rather stable, superficially unremarkable developmental trajectory, comparable across all three languages: NCIs emerge between age 2-3, with NC constructions emerging more or less at the same time as NFAs. When these emerge, they are all generally target-like, with some occasional errors, especially at earlier stages. Postverbal NCIs are abundant, both in child and adult production; this is independently expected, given the overall predominance of postverbal subjects in the languages.

Having established the broad developmental patterns in Romance, I now turn to existing and new Germanic data. I show that comparison across the two language families helps adjudicate between competing approaches to the acquisition of negative indefinites. The empirical discussion proceeds in two parts. First, I examine the error rates observed across the two language groups and the age at which these errors emerge. Second, I consider the presence (or absence) of early isolated negative indefinites – instances where NIs occur on their own before children produce fully grammatical sentences containing them. I contrast these findings with the broad theoretical division outlined in §2, namely between approaches that treat NC as developmentally and/or synchronically ‘default’ and those that take NC systems to be *derivative*, and thus formally more complex

3.4 Comparison with West Germanic

3.4.1 Error rates and error emergence in Germanic vs. Romance production

Hein et al. (2023) and Driemel et al. (2023) present two new, large-scale corpus studies on the acquisition of negative indefinites in German ($N = 43$), English ($N = 7$) in Hein et al., and Dutch ($N = 40$) children (Driemel et al.) on CHILDES. This section, and any West Germanic data cited henceforth, is based on these works. Importantly, I also draw on my own analysis of unpublished data from their raw dataset[14] and, more rarely, from other available CHILDES data. I focus only on the datasets that encompass 'early' stages of acquisition; that is, any corpora whose recordings either begin after age 4-5 or mostly comprise that age period are excluded, to avoid sample skewing. This is because the majority of the recordings of the corpora consulted begin at or before age 2. I exclude cross-sectional corpora and focus on longitudinal trajectories only, as with the Romance corpus study in §3.1.

This led to a sample of $N = 11$ German children (4 children not in Driemel et al., 2023, from the Koch corpus, Koch, 2019), $N = 17$ English children (10 additional children, from the Manchester corpus, Theakston et al., 2001) and $N = 16$ Dutch children. The resulting corpora used were as follows: 6 corpora for German, 3 for English and 6 for Dutch. For data on error production (this section §3.4.1), only Driemel et al.'s (2023) dataset will be reported. The more expansive sample is used to corroborate the pattern reported in §3.4.2.

Broadly, negative indefinites in Germanic children share developmental similarities with Romance: NIs also generally emerge between ages 2-3, on average at 27.2 months. *Nothing* is early-acquired among NIs, though items such as German *kein* (+ DP) and English *no* (+ DP) are also among the most frequently used NIs (Hein et al., 2023). Like Romance, NIs also seem to be preferentially observed postverbally in Germanic (Figure 8; see also Bill et al., 2024), whilst this is not the case in English (Hein et al., 2023), plausibly due to the word-order strictness and strong subject-requirement of the language.

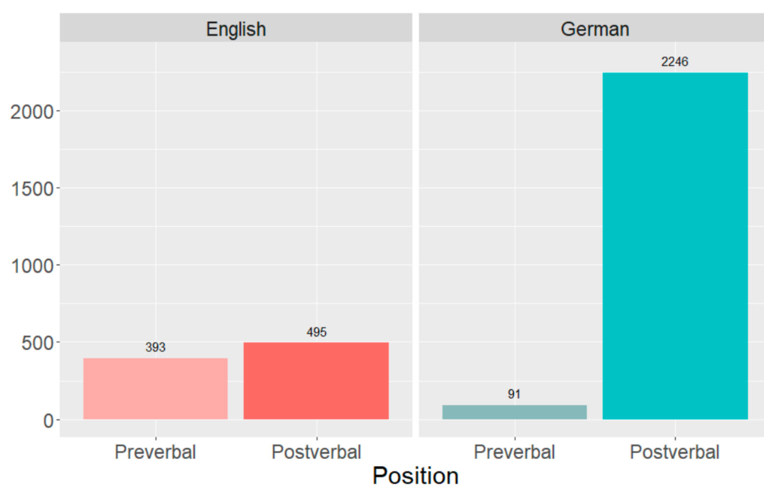


Figure 8: Number of NIs by position (Hein et al., 2023: 10)

[14]With special thanks to Imke Driemel and Johannes Hein for kindly providing me with access to this dataset.

In German, too, a preference for *object* NIs is also found (Bill et al., 2024, 2025), even for NIs which are neutral towards animacy, like *kein* ('no'+DP). This suggests that at least some of the developmental trends in the acquisition of indefinites crosslinguistically share a common source that is possibly not Germanic-specific (see Bill et al., 2024, for example, on one possible explanation for issues with preverbal NIs in German); however, others are reflective of language-specific properties, e.g., word order differences within Germanic and the more complex system of NIs in English (Hein et al., 2023).

As noted in §2, children acquiring a Germanic language have been argued to go through a stage of producing 'NC-like' errors – a pattern that has led to proposals suggesting these errors are due to an underlying NC-like competence, either because the relevant Germanic language (e.g., English) is already synchronically a (hybrid) NC system and/or because children are 'biased' towards NC grammars in some sense (recall §2). This raises the question of how different their error patterns are from Romance children; that is, if child Germanic NC-like productions are comparable to Romance. Note that, although possibly less discussed, the reverse position – that Romance children may pass through a DN-like stage instead – is not ruled out, and it is in fact explicit in other approaches (Zeijlstra, 2004, 2008; with some supporting data in Tagliani et al., 2022). It must then be established which putative non-target-like stage (if any) is favoured by the empirical data in the two families. I will focus on Germanic *production* data so as to be comparable with our Romance production data, leaving discussion of the asymmetry in comprehension to the analysis in §4.

Driemel et al. (2023), drawing on earlier work in Hein et al. (2023), present the following error rates in production for their study on English, German and Dutch children on CHILDES (Table 5, repeated from Table 1). Reported below are errors, all of which correspond to concord errors: the sentential negator accompanies NIs where it should not in the adult target-like system.

Table 5: Production of NIs and NC across English, German and Dutch (Driemel et al., 2023: 8)

	Utterance Count				
Language	**Total**	**with NI**	**with NC**	**Proportion (NC/NI)**	**Peak(s)**
English (all)	328,972	909	184	20.2% (Hein et al., 2023)	34%
English (w/o Adam)	283,399	666	53	8.0%	7.1% & 13.4%
German	338,407	2665	45	1.7% (Hein et al., 2023)	5.9%
Dutch	220,617	857	6	0.7%	3.7%

Examples of such concord errors are given below.

(18) a. English, Mark (2;08)

No tigers **don't** bite you?

b. Adam (3;11)

We **don't** want **no** gas.

c. German, Leo (2;03)

Kein Gewitter kommt **nicht** heute.
no thunderstorm come.3SG not today

'There's no thunderstorms coming today.'

d. Simone (3;07)

Wir haben noch **keine** Zudecke **nich**.
we have.1PL yet no duvet not

'We don't have a duvet yet.'

e. Dutch, Daan (3;00)

En Rosa mag **niet geen** spelletje.
and Rosa may.3SG not no game.DIM

'And Rosa may not play a game.'

f. Diederik (2;10)

Heeft Arnold **niet geen** hamer.
have.3SG Arnold not no hammer

'Arnold doesn't have a hammer.'

(Driemel et al., 2023: 6-7)

Overall, Driemel et al. found a significantly higher proportion of NC for English (184 out of 909 utterances with NIs), than German (45 out of 2665) or Dutch (6 out of 857). The proportion of NC in English decreases substantially with the exclusion of Adam, a child exposed to African American English (AAE). Although his parental input did not contain instances of NC, AAE is a variety with NC, so it is conceivable that his greater proportion of NC is due to reinforcement in his linguistic environment. Even with exclusion of Adam, the error rate for English remains well above German and Dutch – the two sets of languages clearly pattern in developmentally distinct ways and so must arguably be dealt with theoretically at least partly separately. I offer a more detailed discussion of the German/Dutch and English results later in this section and section 4, once the full empirical comparison with Romance has been laid out.

At this point, I would first like to challenge the significance of these error rates as indicative of some putative NC-like stage, by comparing them to the Romance data. Table 6 shows that the Germanic error rates are, in fact, no higher than those of Romance children, at least for children acquiring non-NC varieties (the AAE child Adam being an exception). Both language groups produced similar proportions of errors, albeit very rarely (recall that the Romance errors here involve primarily omission of the negator with postverbal NCIs).

Table 6: Mean proportion of utterances with NCIs and Errors by language

	Utterance Count			
Language	**Total**	**NC structures**	**Errors**	**Proportion (Error/NC)**
Catalan	66,047	54	0	0%
Italian	33,407	54	11	20.4%
Italian (w/o Rosa)	26,794	37	3	8.1%
Spanish	148,296	253	15	5.9%

This suggests that these error rates are commonly found, *irrespective* of the typology of the language being acquired, and in turn casts doubt on the possibility that Germanic children have induced a non-target-like NC system. Importantly, too, the errors in both language families are qualitatively distinct: Romance errors involve *omission* of an otherwise compulsory negator, Germanic errors involve *overproduction* of negators with indefinites. This points to different sources and so suggests that a bias towards NC cannot underlie NI development broadly understood – errors of omission defy a NC bias, which should *encourage* consistent production of the negator instead.

Furthermore, analysis of the NC errors in the Germanic children who displayed a high error count reveals that their putative 'NC-like' system is, at best, highly restricted. The child Leo (Leo corpus) contributes the vast majority of the NC errors in the German sample (33 errors out of 45 in total, 73.3%). Virtually all his errors are restricted to the indefinite *kein* ('no'); only one example is found with *nichts* ('nothing') (*nichts ni(ch)t mehr drin*, lit. 'not nothing more inside'). They do not occur with *niemand* ('nobody') or *niemals/nie* ('never') in Leo.[15] Were these errors produced by a system that is NC-like, we would not expect error patterns to be limited to only one indefinite.

Note, however, that this is, again, *not* the case for the English data: several NC errors with *nobody* and *never* can be found in all of the children who produced errors, bar Eve, and in both preverbal and postverbal contexts.

(19) a. Adam (3;10.15)
I **don't** see **nobody**.

b. Adam (3;11.14)
Nobody won't recognize us.

c. Fraser (2;11.02)
He **doesn't** go on **nobody**'s head.

d. Fraser (3;00.00)
Nobody's **not** drying him.

e. Mark (4;04.16)
Um (.) you [/] you [/] you **shouldn't never** hit someone.

f. Ross (3;11.28)
But I promise I'll **never** hurt **nobody** again.

g. Ross (2;08.17)
No tigers don't bit you?

[15] Errors with *niemand* and *niemals/nie* are only found 3 times across all children studied in Driemel et al. (2023).

h. Eleanor (2;11.06)

He **won't** hurt his head **never**.

That English patterns developmentally differently, even after excluding Adam, is also clear by the proportion of NC errors over age across the three languages, repeated from §2.

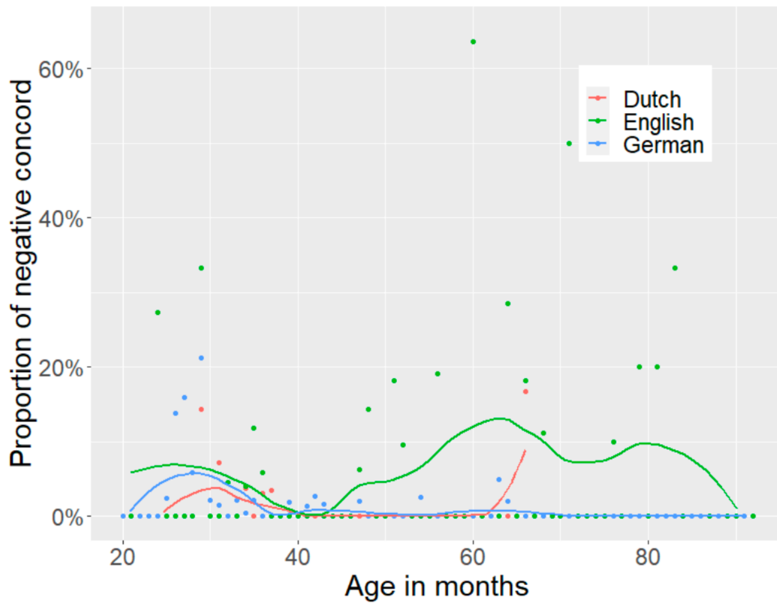


Figure 9: Proportion of NC over age in Dutch, English (excluding Adam) and German (Driemel et al., 2023: 8)

Whilst the majority of NC errors cluster at early age ranges (20-40 months) with peaks of around 5%, English children continue observing higher error peaks beyond month 40 and in some instances beyond 80 months. This arguably raises the need for a separate explanation of the English facts, independently of German and Dutch. Driemel et al. and Hein et al. offer some speculations as to why this may be the case, such as the comparatively more complex systems of Negative Polarity Items (e.g., *anybody*, *anything*) and negative quantifiers in the language (see §2 above). For now, I conclude that the error rates observed for German and Dutch do *not* point to a NC-like stage, they are no higher than error rates for children acquiring Romance languages, who instead produce errors of omission of negators. Nonetheless, the case of English remains developmentally (and typologically) distinct, needing separate scrutiny. I return to English's developmental profile in §4.

The timing at which systematic errors first emerge in the two language families is no less theoretically consequential, in particular relative to the emergence of fully-fledged sentences with negative indefinites or NCIs. Romance children often make omission errors at early stages, as reported in §2.2, sometimes *before* NC structures become established. There is only one child in our dataset that makes systematic recurring errors in her data sample, the Italian child Rosa (already described in Tagliani et al., 2022). The majority of errors involve omission of the negator, although one is also an apparent extension of *nulla* into non-negative contexts (a question in 20a):[16]

(20) a. Rosa (2;04;23, 1.75 MLUw)

CHI: Ha **nulla** in mano?
have.3SG nothing in hand?
'Does (she) have anything in hand?'

MOT: Non c'ha nulla in mano amore, nulla
not have.3SG nothing in hand, love, nothing
'(She) doesn't have anything in hand, love, anything'

b. Rosa (2;07.26, 2.84 MLUw)

Questo fa **nulla**, guarda
this do.3SG nothing look
'This is doing nothing, look.'

As Tagliani et al. report, from 1;11-2;02 Rosa uses the NCI *nulla* ('nothing') grammatically as a negative fragment answer and produced two instances of NC from 2;03-2;06. Omission errors are still predominant until 2;10, making up 85% of her utterances with NCIs. The first instance of correct NC is at 2;07; this coincides with the time when Rosa begins to use *non* consistently as the sentential negator, and abandons the anaphoric negator (as in 'no+VP') as the primary means to encode sentential negation (Tagliani et al., 2022: 17-18).

A similar development is observed to some extent in Martina (Italian) who has not yet acquired NC and produces an error at 2;03.01 (*c'è nessuno*, 'There's nobody'). Grammatical NC is not observed until two weeks later (2;03.33). Carla (Spanish) only produces two negative indefinites: the first one at 3;00 omits the compulsory negator; this is later 'corrected' at 3;02, where the negator is overtly produced.

Although not categorical due to sparsity of errors in the first instance, a trend can be obtained when the data tabularised and visualised (Table 7). The table reports the age in which the first error

[16] Although, as noted by Tagliani et al. (2022: fn. 8), the utterance in (20a) is grammatically accepted in some Tuscan varieties to express semantic negation. This could therefore be due to dialectal variation rather than an omission error.

was found and the age range in which NCIs are found in (grammatical) fully-fledged sentences, for ease of comparison and contextualisation. The children are ordered by first emergence of NCIs. Only children who produced both errors and NCIs in fully-fledged sentences are included. I do not include data for NFAs here, on the grounds that errors in Romance can only be observed if a verb is present.

Table 7: Overview of Romance children producing NCIs and errors

Child	Language	Type	Age Range with NC	First Error	*N* of Errors
Magín	Spanish	Monolingual	1;09.27–2;04.25	1;09.27	1
Elisa	Italian	Monolingual	1;11.19–2;01.06	1;11.19	2
Irene1	Spanish	Bilingual	1;11.01–3;02.19	3;00.24	1
Martina	**Italian**	**Monolingual**	**2;03.22**	**2;03.01**	**1**
Simon	Spanish	Bilingual	2;05.26–3;05.25	2;10.02	2
Mar	Spanish	Bilingual	2;06.13–5;00.17	2;09.20	1
CHI2	Spanish	Bilingual	2;07.11–6;05.29	3;10.00	4
Rosa	**Italian**	**Monolingual**	**2;07.26–3;03.23**	**2;02.11**	**8**
CHI1	Spanish	Bilingual	2;10.21–6;02.09	2;10.21	2
Emilio	Spanish	Monolingual	2;11.08	2;11.08	1
Carla	**Spanish**	**Bilingual**	**3;02.23**	**3;00.16**	**1**
Average onset (months)			29.27	31.91	23

We can note, then, that errors generally emerge on average within 1-3 months of the first NC structures, and in three children *before* grammatical NCIs are produced (bolded in the table). Errors are an 'early' phenomenon within the development of NCIs in Romance.[17]

Contrast now the Romance data with Table 8, which compares the age of emergence of errors

[17]This is especially true of monolinguals (average 27.4 months for NC vs. 26.4 for Errors). Bilinguals appear to develop errors at a later stage (31.9 vs. 36.5 months), though still earlier than Germanic children (see Table 8). Note, however, that exceptions to this trend nonetheless exist (see, e.g., Carla in the table). The sample of children with errors for monolinguals vs. bilinguals is too small to draw conclusions, but it is an important avenue for future work to check if this discrepancy generalises to a broader sample and if differences are observed in bilinguals of different language pairs.

Table 8: Overview of Germanic children producing NIs and errors in Driemel et al. (2023)

Child	**Language**	**Age Range with NIs**	**First Error**	***N* of Errors**
Corinna	German	1;08.11-7;06.10	2;11.24	3
Eve	English	1;09.00-2;03.00	2;00.00	2
Caroline	German	1;09.03-3;04.01	2;09.08	2
Fraser	English	2;00.27-3;01.03	2;00.27	16
Eleanor	English	2;00.23-3;00.30	2;11.06	1
Leo	German	2;01.06-4;11.05	2;02.02	33
Simone	German	2;01.16-4;00.06	3;07.07	1
Daan	Dutch	2;02.02-3;03.30	2;05.11	3
Mark	English	2;04.01-5;08.28	4;00.22	15
Laura	Dutch	2;04.01-5;06.12	5;06.12	1
Sebastian	German	2;06.17-7;05.11	2;10.21	3
Cosima	German	2;07.22-7;02.22	2;09.03	2
Ross	English	2;08.05-7;08.02	2;08.17	19
Matthijs	Dutch	2;09.26-3;07.02	2;09.26	1
Adam	English	2;11.13-5;02.12	3;04.01	132
Average onset (months)		26.87	35.67	234

across the three Germanic languages and the full dataset in Driemel et al. (2023). Only children who produced errors are included here, as above. Errors in Germanic emerge well after Romance errors, 9 months later on average.[18] But more significantly, given the individual variation in errors produced, there is no child that produces NC errors *before* grammatical uses of NIs are made, despite the bigger and denser sample.

The difference between Errors vs. full sentences is only significant in Germanic:

[18] I exclude uses of *no* in English, followed by a DP, that are ambiguous between the negative indefinite or the early-stage (non-target-like) use of the anaphoric negator in place of the sentential negator *not* (Klima & Bellugi, 1966).

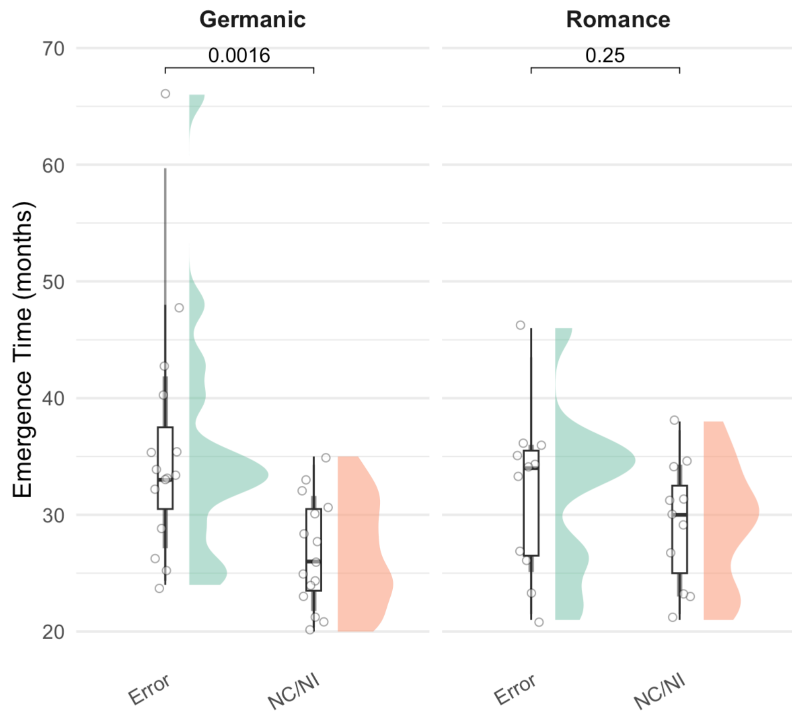


Figure 10: Emergence of Errors vs. NC/DN by Language Group

This suggests NC errors are a phenomenon in Germanic that consistently *follows* the grammatical use of negative indefinites in full sentences. This is corroborated by analysing the profile of the children who display a systematic stage with errors (with 10+ errors): Leo (German), Fraser, Mark, Ross and Adam (English). The existence of a systematic stage of overuse of concord is only largely clear-cut in some of the English children: Fraser produces the first grammatical indefinite at 2;00. A concord error is found in the same file, but errors are not frequent until 2;05; these errors are sporadically produced until age 3;00. The siblings Mark and Ross share a similar trajectory, with negative indefinites appearing first grammatically at 2;04 and 2;08, respectively, before producing several concord errors from ages 4;04 and 4;00, respectively. Finally, Adam is a transparent case of later use of concord, plausibly facilitated by his input or environment: his first negative indefinite appears at age 2;11. Despite the likely reinforcement in the input, he does not produce concord structures in English until 3;04; these then become very frequent thereafter (131 occurrences) until the end of the recordings.

Only one German child, Leo, consistently produces instances of apparent NC, though these are often in verb-less sentences. He produces the first sentences with a grammatical (non-concord) negative indefinite at 2;01.06. A couple of early concord errors are found without a verb at 2;02 (e.g., *Keine Enten nicht da!*, no.NOM.PL duck.PL not there, 'No ducks there!'), but these do not become more frequent and include inflected verbs until 2;03.

We then see that consistent omission errors in Romance often occur *before* the systematic production of NC structures or right during the first occurrences of NC. Later production of errors is nonetheless attested, but often in children who only produce one error in total. In Germanic, on the other hand, NC-like errors consistently emerge after the establishment of negative indefinites in fully-fledged sentences, often half a year or a year after the latter. They are never produced *before* NIs are uttered grammatically in a sentence. This holds for all children, irrespective of whether production of concord errors is commonplace in them or not. Again, Adam is the child that systematically produces more errors. Nonetheless, he observes the same pattern as the other children: a stage with a DN system, only allowing NIs to appear on their own (from 2;11), precedes his stage with consistent NC errors (from 3;04).

This section has emphasised two points: (i) Germanic error rates are not significantly higher than those of Romance children, they are comparable; (ii) Romance and Germanic (particularly English) pattern oppositely as regards age of emergence of errors. I argue in §4 that approaches assuming NC is developmentally primary do not foresee these two results (see also discussion in Zeijlstra, 2022: 134ff). This (tendential) crosslinguistic difference is, however, expected under a theory where DN is featurally simpler, such as Zeijlstra (2004).

Before doing so, I first turn to a second result to reinforce my argument: Germanic children frequently produce (non-target-like) negative indefinites in isolation early on to indicate rejection, disapproval or comparable negative meanings, without being negative fragment answers to a previous wh-question. This behaviour is systematically unattested in the Romance children.

3.4.2 Isolated uses of negative indefinites: a typologically-conditioned asymmetry

A further developmental difference between Romance and Germanic emerges once the very first uses of negative indefinites are examined. Recall that Romance children produce NFAs approximately around the same time as NC constructions (Table 7). The pattern appears to recur for Germanic children (where 'NI_sent' below refers to grammatical full sentences containing a NI). Table 9 and the data reported henceforth makes use of the dataset in Driemel et al. (2023) but also supplementary data collection (14 additional children), as outlined in §3.4.

Structure	***N***	**Mean**	**Median**	**SD**	**Min**	**Max**
NI_sent	38	29.39	28.5	4.88	21	48
NFA	34	31.09	30	5.34	24	51
Isolated NI	29	27.21	27	4.03	22	37

Table 9: Descriptive statistics for emergence time (months) by Structure in Germanic.

Whilst the emergence of grammatical uses of negative indefinites (e.g., in full sentences or in NFAs) is comparable across the two languages, several Germanic children produce earlier instances of negative indefinites before their grammatical uses are acquired: these involve non-target-like uses of *isolated* negative indefinites that do not correspond to negative fragment answers. I term these *isolated* uses of negative indefinites, to avoid confusion with the term fragment answers. (21) gives two examples from each language:

(21) a. English, Gail (1;11.27, 1.64 MLUw)

*CHI In there.

*MOT Pour it in then.

*CHI **Nothing**.

*MOT Nothing in there?

b. Adam (2;03.04, 2.09 MLUw)

*MOT Do you remember David?

*CHI No. **Never** David.

c. German, Leo (2;01.19, 1.27 MLUw)

Doch **keinen**. **Keine** Katze da!
but none no cat there

'But none. No cat there!' (CHI looks into the courtyard where he has seen a cat before but doesn't see one today.)

d. Kerstin (1;11.20, 1.53 MLUw)

*MAX Du telefonierst aber angestrengt!
you telephone.2SG but strained
'You're shouting on the phone!'

*MUT Hier! Wart!
here wait.IMP
'Here! Wait!'

*CHI Oma **keiner** da.
grandma nobody there
'Grandma, nobody there.' (alternatively, 'Grandma isn't there', 'At Grandma's place, there's nobody')

*MUT Sagste mal mit der Edith reden.
say.IMP-you once with the Edith talk
'Tell her you should talk to Edith.'

e. Dutch, Abel (2;10.00, 2.81 MLUw)

*JEA Is lekker, he?
is tasty he
'It's tasty, huh?'

*CHI **Niks** papa.
nothing dad
'Nothing, dad.' (≈ 'It isn't, dad!')

f. Peter (1;11.25, 1.77 MLUw)

*FRA Zullen we hier mee tekenen? Zullen wij even gaan tekenen ?
should.1PL we here also draw.INF should.1PL we even go draw.INF
'Should we also draw here? Should we go and draw?'

*CHI **Geen**.
no
'No/none' (≈ No, I don't want to).

*FRA Geen? Wil je die niet?
no want.2PL you that not
'No? Don't you want that?'

(21b, 21e, 21f) shows how Germanic children produce negative indefinites in isolation, with a reading resembling rejection, disagreement, disapproval or comparable. Other examples (21a, 21c, 21d) appear to suggest a full proposition was intended, despite absence of a verb. In all cases, the flavour of the contribution of these negative indefinites is clearly negative and, crucially, they are not fragment responses to a previous wh-question. In some cases, no question is uttered by other interlocutors at all. Although *nothing* predominates (being the first NI to emerge for most children), examples are found with both *no*+DP (especially in German and Dutch) and also *nobody* and *never* (see 21d), though the latter are rarer. This shows that the isolated use of NIs to indicate negation is not item-specific.

In some children, these isolated productions are rather common at particularly early stages: Gail produces her first isolated NI at age 23 months, with uses of NIs in full sentences not emerging until 32 months. She frequently produces *nothing* as an isolated NI denoting negation or as truncated copula sentences (e.g., 'nothing in there') from the first file (1;11) up until 2;01 (8 instances), with none of these examples being answers to wh-questions. NFAs emerge subsequently at 2;02, from which point isolated NIs disappear. Adam's first isolated NI also comes in early (27 months), before full sentences with NIs are observed at 35 months. Peter (Dutch) likewise produces frequent isolated NIs from 23 months. These productions are extremely frequent, with *geen* consistently being used as a way to negate or refuse previous statements (*geen* is used in isolation primarily, even though in the adult grammar it is generally followed by a DP, like English 'no'+DP). Earlier development of isolated NIs is also observed above in Leo above (21c; 24 vs. 25 months) and Kerstin (23 vs. 27 months). In total, 18 children produce isolated NIs *before* they appear in complete utterances; this is out of a sample of 42 Germanic children, 29 of whom show isolated uses of NIs.

Figure 11 shows Kaplan–Meier survival curves representing the cumulative emergence of the three syntactic structure types – Full sentences (here 'NI_sent'), Isolated uses, and Negative Fragment Answers (NFA) – as a function of age. Each curve illustrates the proportion of participants who had produced the respective structure by a given age.

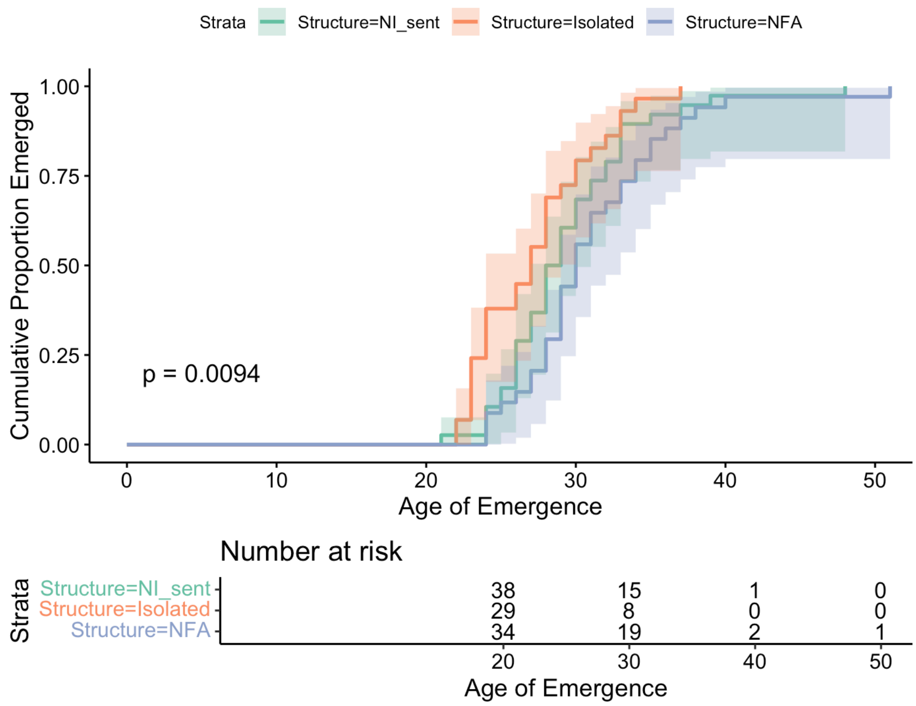


Figure 11: Cumulative Emergence of Structures using Survival Analysis

Emergence of isolated fragments occurred earlier than NI_sent and NFA structures, with a steeper slope in Figure 11. A log-rank test revealed a significant difference in emergence distributions across the three structure types ($p = 0.0094$). The risk table below the plot indicates the number of children so-called 'at risk' (i.e., those who had not yet produced the structure) at 10-month age intervals.

What is remarkable here is that these isolated indefinites are completely absent in the Romance data: no comparable examples are found at early stages, and the few uses of NFAs or isolated NCIs that are used in more 'non-canonical' ways are fully grammatical in the adult language – for instance, as NFAs with discourse functions produced at relatively late stages:

(22) a. Irene (2;09.11, 3.65 MLUw)

*MOT ¿Cómo se la vas a cuidar?
how CL.IO= CL.DO= go.2SG to take.care.of.INF
'How are you going to take care of it (for her)?

*CHI **Nada** (.) No tiene nada.
nothing not have.3SG nothing
'Nothing (≈ I won't have to take care of it). She hasn't hurt herself.'

b. Irene (2;11.27, 4.0 MLUw)

*MOT ¿Y a Cuqui qué le vamos a decir?
and to Cuqui what CL.IO= go.1PL to say.INF
'And Cuqui, what are we going to say to her?'

*CHI **Nada**, que fuimos a ver a los Arenal.
nothing that go.PST.PFV.1PL to see.INF DOM the Arenal
'Nothing (≈ I won't say much, only that...), that we went to see the Arenal family.'

These clearly differ both functionally and distributionally from the early-stage isolated indefinites observed in the Germanic data, and are always grammatically used when following a question by another interlocutor.

This contrast is important, given the almost-identical number of children studied across the two languages (43 Romance vs. 42 Germanic children). The complete absence of isolated indefinites in the Romance data prior to the acquisition of NCIs in adult-like constructions suggests that the asymmetry is not a result of mere sampling bias, but instead reflects a deeper structural or developmental difference between the language groups.[19] I argue in §4 that it in fact 'tracks' the proposed semantic nature of the NIs in the two language groups: Germanic NIs are standardly treated as inherently semantically negative (negative quantifiers) and so can convey negation *on their own.* I follow accounts that treat Romance NCIs as semantically non-negative. They generally cannot appear in isolation, requiring therefore an overt (or covert) licensor.

4 Discussion and analysis

We began the paper by introducing a potential point of contention between (some) theoretical approaches to Negative Concord and the empirical data from the acquisition of negative indefinites

[19] I thank Esther Rinke (p.c.) for raising another potential interpretation of these isolated indefinites: namely, that they reflect full intended utterances whose verb has been omitted. Germanic children are known to stabilise the use of finite verbal morphology later than Romance children (viz., the Root Infinitive stage), and so might (putatively) drop verbs more frequently. In this view, Germanic isolated indefinites would be analogous to adult-like structures with a verb, making them structurally comparable to the omission ('DN-like') errors produced by Romance children (where an NCI is used in absence of an accompanying negator). I do not adopt this analysis for the present data: not all examples in (21) are straightforwardly utterances with elided verbs (see, e.g., 21e–21f). Moreover, the absence of overt inflectional morphology in Germanic development is not itself grounds to expect this omission to also extend to *whole* verbs (not just their overt inflection). Without independent support for this latter assumption, there is no clear reason to predict a difference between Germanic and Romance children in their use of isolated indefinites.

in Germanic: recent work has argued for a competence root to early errors in the production and comprehension of Germanic NIs; that is, that Germanic children go through an initial NC stage, before converging to their target-like DN grammar. Conversely, several synchronic analyses suggest that NC is *derivative* (and so acquisitionally harder) than DN systems (e.g., Zeijlstra, 2004; Deal, 2022). A challenge remains as to how make these empirical and theoretical results compatible in a theoretically parsimonious way.

The goals of this paper were two-fold. The first were *empirical* and *descriptive*: I introduced the first overview of the acquisition of Negative Concord Items and NC constructions in Catalan and Spanish. I then turned to a *comparative* and *theoretical* analysis of the Romance results (including Italian): I argued that the results of the comparative study of the acquisition of NI in Romance vs. Germanic have high scientific potential – specifically, they shed direct light on the debate regarding the development status of NC systems.

At a broad level, Romance and Germanic behave largely similarly: NIs emerge at comparable ages, if anything earlier in Germanic.[20] Error rates are comparable across the two families, although NIs are used less frequently in Romance. This overall convergence is itself informative: if Germanic children were entertaining NC-like systems, their error rates should outstrip those observed in Romance. The observed rates alone therefore do not warrant positing a NC stage in Germanic. More generally, children are known to both overmark single interpretations (Rowland, 2007; Ambridge et al., 2015; Guasti et al., 2023), and also to omit markings, even when their systems are otherwise largely adult-like. Occasional errors of the kind in Germanic and Romance are thus not unexpected.

At the same time, a closer look at production reveals clear language-specific differences, concerning (i) the type of errors, (ii) their age of emergence, and (iii) unattested error types. Romance children primarily produce (rare) omission of the negator. Crucially, these errors arise early, in some cases before the first grammatical uses of NCIs. In Germanic, by contrast, errors tend to emerge later. Finally, overproduction of negators with preverbal NCIs is unattested in Romance: errors are exclusively postverbal across all three languages.

I argue that these results are sufficient to disfavour a 'NC-default' account. Given comparable error rates, such an account fails to explain why Romance children sometimes 'struggle' to produce

[20] Cf. Katsos et al. (2016), who speculate that acquiring a Negative Concord language may facilitate earlier acquisition of negative indefinites.

NC constructions – a surprising outcome if NC were the default hypothesis. Errors of omission amount to *overcompression* errors under a Meaning First approach and would require *ad hoc* deletion rules that must later be retracted from. This raises learnability concerns and runs counter to the Undercompression Hypothesis of the approach. The cross-family age asymmetry also receives no treatment: the earlier emergence of errors in Romance cannot be reduced to inter-child variation, as it systematically tracks language family. It must then receive an appropriate L1-based account. Likewise, the absence of errors with preverbal NCIs in Romance exposes a limitation of existing accounts of NC errors in Germanic: they do not make a commitment regarding which 'kinds' of NC-errors will be observed: for example, strict-NC-like errors (negator overproduced in all contexts), non-strict-NC-like errors (overproduction in postverbal contexts only) or both. Both error types are crucially produced in English children (19), a language which synchronically permits these NC constructions in some varieties, but errors with preverbal NCIs are absent in the Romance production. This contrast suggests that children are sensitive early on to the typological properties of their target system: Romance children, in particular, appear to respect the non-strict nature of Catalan, Spanish, and Italian from the onset of NCI production. Under a NC-default account, we thus require an independent explanation as to why the skewing in types of NC-errors crosslinguistically manifests itself the way it does, leading to a less parsimonious account.

Further, I then showed how the earliest productions of NIs across Romance and Germanic reveal an asymmetry: Germanic children frequently produce isolated NIs with negative force (e.g., to express rejection), in non-target-like ways. This behaviour is systematically absent in the Romance children studied. The contrast again aligns with the properties of the target systems (Double Negation vs. NC), indicating that children differentiate between them from early on. Taken together, these findings argue against a uniform NC-like initial state in the acquisition of NIs.

Overall, a comparative perspective on NI-acquisition then rules out particularly strong versions of the putative NC-stage (e.g., Thornton et al., 2016; Sano et al., 2009) and would force the postulation of additional machinery for the Romance data under other, more lenient accounts (e.g., Meaning First; Driemel et al., 2023).

In particular, I take the key conclusion of the dataset to be that NC and DN systems differ in fundamental ways in their developmental trajectories. This asymmetry calls for a formal account that can derive the observed bipartition. I suggest that the most relevant distinction lies at the level of the *semantics* of NIs crosslinguistically, and propose that the approaches best suited to capturing the data are those that treat NCIs (but not negative quantifiers) as *non-negative*. Specifically, I adopt Zeijlstra's (2004, 2008) feature- and Agree-based approach to implement my analysis. My analytical choice is motivated by two considerations. First, unlike many of the accounts reviewed in §2, Zeijlstra (2004) posits an explicit formal division between NC and DN systems grounded in the semantic vs. syntactic status of negative features. Second, as a feature-based approach, it lends itself naturally to acquisition-based predictions within featural and parametric frameworks (e.g., Roberts & Roussou, 2003; Zeijlstra, 2008; Biberauer, 2019). On this view, I show that the patterns observed in the data follow independently, including the absence of isolated NIs in Romance and the 'special' behaviour of English within the Germanic group.

Recall, as summarised in §2, that Zeijlstra (2004, *et seq.*) assumes NCIs in Romance are non-negative items, endowed with an uninterpretable [uNeg] which must be valued by a matching [iNeg] (e.g., a sentential negator) to lead to grammaticality. German, Dutch and (Standard) English, in contrast, have truly negative NIs, which are analysed as negative quantifiers (with a semantic [Neg] feature). Zeijlstra presents an acquisition pathway in which these features will be acquired. All things equal, the child will not postulate [iNeg/uNeg] pairs – these are *formal*, *syntactic* features which require triggering by the input – and will default to analysing NIs as semantically negative, i.e., specified with a semantic (rather than syntactic) negative feature. Negation is never formalised in such a scenario. This effectively makes Double Negation systems the default outcome in acquisition.

How, then, can the child postulate a syntactically active [iNeg], which values [uNeg], leading to NC systems? The learnability logic is captured by Zeijlstra's (2008: 153) *Flexible Formal Feature Hypothesis* or FFFH: whenever there is 'doubling' with respect to a semantic feature F – i.e., a given feature or dependency is manifested more than once –, this will lead the child to postulate [iF/uF] pairs to syntactically encode the relevant dependency:

(23) **Flexible Formal Feature Hypothesis (FFFH)**

a. If and only if there are doubling effects with respect to a semantic operator OP_F in the language input, all features of F are formal feature [i/uF].

b. If there are no doubling effects with respect to a semantic operator OP_F in the language input, all features of F are semantic features ([F]).

Doubling with respect to negation is readily detectable: all else being equal, two instances of semantic negation should cancel each other out. When this does not occur, a doubling effect arises, providing evidence for [iNeg]/[uNeg] features on the relevant items. This is the treatment used for Negative Concord in this system (recall §2; see also Zeijlstra, 2004: ch. 8). This approach yields clear predictions for acquisition. Since Double Negation is the default, Germanic systems, being DN, are expected to show low error rates. Where systematic errors do arise, the account predicts their age of emergence to be exactly in the direction observed here. In Romance, children will by default treat NCIs as independently negative; this can then give rise to early DN-like errors (e.g., omission of the negator), before or around the stage at which [uNeg] is postulated for NCIs. In Germanic, by contrast, children are expected to remain within a DN system unless the input motivates a shift towards NC. As a result, NC-like errors should emerge only at a later stage, once NIs and DN are already in place, and only in contexts where the input provides evidence for NC.

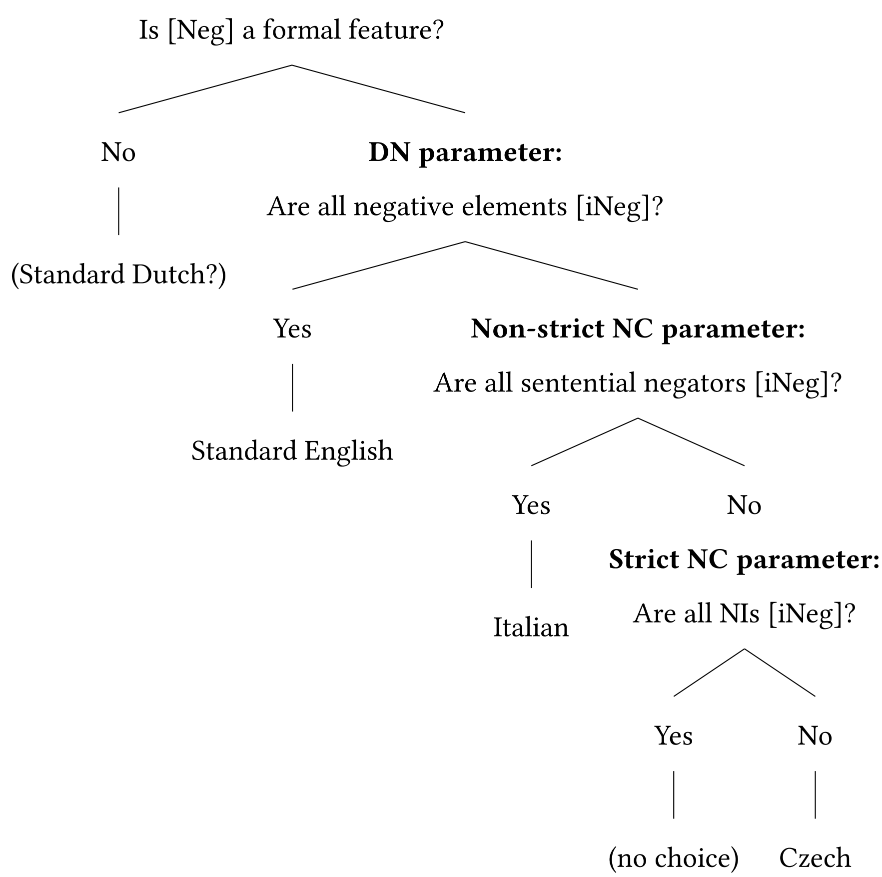
The high frequency of errors in *only* the English child data clearly exemplify this latter point (recall Table 5). They are consistent with English's proposed 'hybrid' status regarding NC/DN: for example, Blanchette (2015), based on acceptability judgment experiments, shows that speakers of Standard English systematically allow NC interpretations for sentences that are prescriptively associated with DN, suggesting that NC readings are grammatically available. Similarly, Zeijlstra (2004) argues that contracted negation (*n't*) is a head rather than a phrasal negator, and can therefore participate in NC-like configurations despite the language not being uniformly NC (see also Zeijlstra, 2022). By contrast, such errors are extremely rare in German and Dutch, which lack NC properties synchronically. There is thus a qualitative difference in development, which reflects the input: NC-like behaviour is virtually absent in German and Dutch, but attested in English at later stages. This in turn raises the question of whether at least some of the very sparse errors observed in German and Dutch (and indeed Romance) reflect performance noise rather than a genuine developmental stage.

The absence of errors with preverbal NCIs in our Romance data is also predicted under a parametric approach to acquisition (e.g., Zeijlstra, 2004; Biberauer & Zeijlstra, 2012; Biberauer & Roberts, 2015). First, there is no relevant input that would cue a strict NC grammar. Second, from an acquisitional perspective, strict NC is featurally more complex than non-strict NC. Together, these factors make strict NC an unlikely hypothesis for children exposed to non-strict NC systems, thereby accounting for the absence of errors involving preverbal NCIs.

Zeijlstra (2004) analyses strict NC (e.g., Czech) as involving two sets of uninterpretable negative features ([uNeg]): one on the sentential negator and another on the NCI. Crucially, neither of these elements is itself semantically negative. Instead, both must enter into an Agree relation with a covert negative operator bearing an interpretable feature ([iNeg]), which provides the semantic negation. Because both the negator and the NCI depend on this abstract licenser, NCIs can co-occur with negators in both preverbal and postverbal positions. In non-strict NC languages such as Italian, by contrast, the sentential negator is itself the bearer of the interpretable negative feature ([iNeg]) and thus functions as the licenser. NCIs carry [uNeg] features and must be licensed by this overt negator. As a result, NCIs cannot surface freely in all positions; in particular, no NCI can appear to the left of the negator and outscope it. This contrasts with strict NC languages, where the distribution of NCIs is not constrained in the same way, given that neither the negator nor the NCI independently contributes negation.

The reasoning above is illustrated in the parametric decision tree below (from Biberauer, 2015).

(24)



A Standard-English-acquiring child can generalise [iNeg] to all negative elements (indefinites and negators), leading to a Double Negation system. Acquirers faced with non-strict NC, however, must depart from this featurally simpler system in favour of one in which negative indefinites are non-negative, i.e., they bear [uNeg]. The existence of multiple ‘negative’ elements yielding a single negation reading (e.g., Spanish ***no*** *come* ***nada***) provides the crucial cue for this reanalysis, restricting the distribution of [iNeg] to sentential negators only. Strict NC represents a further step in this process of restriction. In such systems, doubling is observed both preverbally and postverbally, and the distribution of NCIs and negators forces children to abandon the hypothesis that sentential negators bear [iNeg]. Instead, they must analyse both negators and NCIs as bearing [uNeg], with semantic negation contributed by a (structurally higher) covert [iNeg] operator that licenses both via Agree. Assuming that children only postulate features when required (Roberts & Roussou, 2003), these systems can be ordered along a cline of acquisitional complexity: Double Negation, Non-strict NC, and Strict NC, in increasing order of featural complexity.

I also propose that the absence of isolated NIs at early stages in Romance directly correlates with the semantic properties of NIs in Romance vs. Germanic. Germanic children appear to have latched on to the fact that NIs in the three languages can convey negation independently. This potentially gives them an expressive resource at early stages: even with a limited vocabulary, they can deploy isolated NIs to articulate strong negative stances – such as refusal, denial, or rejection – resulting in a relatively liberal use of these forms, even when such uses are non-target-like. In other words, Germanic children have access to elements that are unambiguously negative and can therefore be recruited to express emphatic or affective meanings.

In contrast, Romance children develop early on an awareness that NCIs require ‘licensing’ and cannot appear on their own (bar in Negative Fragment Answers). As a result, they lack a comparable expressive resource: NCIs do not encode negation. This limits their availability for the early expression of strong negative meanings. This contrast is not expected under the other accounts considered. Crucially, then, this developmental data can help shed light on the plausibility of synchronic analyses of NIs in the two language groups, with the present data arguably favouring an account assuming non-negative NCIs (*pace* Zanuttini, 1991, and others).

At first glance, the reasoning here may appear circular. If the link between the syntax-semantics of NIs and the (non-)availability of ‘isolated’ NIs is correct, the absence of such uses in Romance suggests that children are already aware that NIs are *not* negative quantifiers – in turn implying early knowledge of NC. This appears to conflict with Zeijlstra (2004), who proposes that Romance acquirers initially may pass through a DN-like stage in which NIs are treated as negative. However, this tension is only apparent. Nothing in the account precludes the early acquisition of a [uF] (such as [uNeg]), provided that the input offers sufficiently robust cues.[21] In this respect, it is well established that the category of negation is acquired early (see Thornton, 2020). Convergence to an NC grammar in Romance is indeed strikingly rapid: very few errors are attested for most children, despite the need to postulate [uNeg]. This suggests that [uNeg] is introduced early in the grammar, plausibly facilitated by the transparent doubling patterns present in the input. Granting this, it is plausible that Romance children quickly converge on an analysis in which NIs require licensing, thereby avoiding isolated uses. At the same time, the available production data may not be sufficiently rich to capture transient non-target-like stages. Experimental and elicitation studies could shed further light on this issue. In particular, data from children who frequently produce omission errors may reveal whether isolated NIs or more robust DN-like stages ever arise. In short, I take the broad longitudinal picture above to be consistent with Zeijlstra (2004) and the acquisition-based approach proposed.

Finally, I attribute the (robust) asymmetry between comprehension and production to a combination of factors. Following Hein et al. (2023) and Driemel et al. (2023), increased processing demands in negation comprehension, especially for double negation readings, are likely to play a role (see Horn, 1989). This may reflect processing limitations, potentially compounded by difficulties in cancelling double negation under time pressure, as argued for Mandarin Chinese by Jou (1988), Zhou et al. (2014), and Hu et al. (2024). Pragmatic factors may also contribute, given the restricted distribution of double negation interpretations in adult grammars (see Driemel et al., 2023). Crucially, mismatches between production and comprehension are independently attested in other complex constructions. For instance, production has been shown to precede comprehension in the acquisition of object relative clauses and certain focus-particle constructions (Friedmann et al., 2009; Müller et al., 2011; Hüttner et al., 2004). However, Illingworth et al. (2025) show that the misinterpretation of *any* as an NCI arises even in the absence of double negation, which argues against a purely processing-based account. I therefore leave open whether the comprehension patterns are entirely performance-based or partly reflect grammatical (e.g., semantico-pragmatic) factors. What matters for present purposes is that such comprehension-production mismatches

[21] This is particularly credible if we consider the preference for postverbal subjects in the Romance languages studied (see §2-3). Postverbal NCIs, and their accompanying negators, create straightforward doubling effects, in contrast to preverbal NCIs in non-strict NC.

are not unusual in syntactic development.

There are a range of additional future directions for the present work and analysis. A clear avenue of future work involves investigating the acquisition of other NC systems, such as children acquiring strict NC languages. Afrikaans (Biberauer & Zeijlstra, 2012) would be another system particularly relevant to test, since it shares patterns with both German and Romance negation systems. It is the reverse of non-strict NC: the language is argued to be a NC system that has semantically negative neg-words ([iNeg]), but semantically non-negative ([uNeg]) negative markers. Future work should test how Zeijlstra's predictions work out for this language: in particular, we would anticipate NIs to behave like those in DN languages (the Germanic children here), but we may anticipate some struggles with the licensing requirement on negators and the two separate *nie*s (a negator and a concord item) that the language has. Alternatively, it is possible that the children show early awareness of this system, just like Romance children (for data showing some early use of concord and NCIs, see Biberauer & van Heukelum, 2025a,b). A final avenue of research would be the effect of bilingualism on the acquisition of NC dependencies, particularly in children acquiring typologically distinct languages. Although data from bilinguals was collected in this study, no straightforward differences were observed with monolinguals.

5 Conclusion

This paper has harnessed comparative developmental data from Catalan and Spanish to inform theoretical debates in the acquisition of NC. I introduced the first study on the acquisition of Catalan and Spanish NCIs and compared it, along with Italian data, to the development of NIs in Germanic. This comparison reveals significant convergences, but also important microparametric differences. On the one hand, I argued that there is no clear support for a NC-like stage in Germanic, at least for production – Germanic error rates are no higher than those produced by Romance children. At the same time, I showed that the error patterns are instead predicted by Zeijlstra's (2004, 2022) Agree-based and non-negative approach to NCIs, including the kinds of errors observed per language, the age at which those emerged and any unattested (but nonetheless logically possible) errors. Finally, I presented a novel pattern: Germanic children sometimes produce NIs in non-target-like, isolated contexts to convey a negative meaning. Romance children systematically do not display this trend and instead produce NCIs in target-like contexts (NC structures or fragment answers) virtually from the start. I related this to the formal nature of NIs in the two families: negative quantifiers in the case of Germanic vs. non-semantically negative indefinites in Romance. Overall, comparative developmental data can help clarify competing analyses of a given phenomenon, and the crosslinguistic acquisition of NIs provides a useful case in point.

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